

SFT V11.12: Entanglement as Shared Boundary Structure

Decoherence, Entropy, and the Architecture of Spacetime

Shaun Fosmark

May 2026

Abstract

The V11.2 quantum foundations program established a two-layer architecture: local dent mechanics for single-particle phenomena, and boundary-level correlation grammar for entanglement. The local layer reproduces interference, decoherence, uncertainty, and the de Broglie relation. The boundary layer reproduces Bell-violating correlations with perfect no-signaling. What V11.2 left open was the physical identity of the boundary correlation. This note proposes that identity.

The proposal: entanglement is shared boundary structure. When two particles are created together, the creation event is written to the holographic boundary as a single entry — one history increment serving two spatial locations. The particles separate in space but their boundary root remains unified. The correlations between them are not carried through space and not pre-determined at creation. They are carried by the shared root, which constrains both particles' measurement outcomes until the root is resolved into independent entries.

A four-gate computational ladder supports the algebraic spine of this proposal. Gate 1A shows that total-spin closure of a shared root uniquely yields the singlet. Gate 1B1 shows that a scalar two-spinor boundary write uniquely selects the antisymmetric epsilon tensor. Gate 1B2 shows that isotropic $SU(2)$ twirling projects arbitrary two-leg writes onto the epsilon component. Gate 1C shows that source quantum numbers select singlet versus triplet records, preventing overclaiming. All four gates pass to machine precision. The singlet kernel $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$ and the Tsirelson bound $S_{\text{CHSH}} = 2\sqrt{2}$ emerge from the derived root without being imported.

If this identification is correct, decoherence is the progressive resolution of shared entries into independent entries, entropy counts the independent entries, the arrow of time is the direction of increasing independence, gravity is dense shared history resisting separation, and expansion is the global dilution of shared history. These are not five separate claims. They are five projections of one mechanism: the universe progressively resolving its original unity into independent components, one tick at a time.

This note is a conceptual seed with initial computational support, not a complete derivation. It organizes a speculative framework with explicit connections to existing V11 results and explicit gates for future development.

Contents

1	The Two-Layer Architecture	3
2	The Proposal: Shared Boundary Roots	3

3	Why This Survives Bell's Theorem	4
4	Decoherence as Root Resolution	4
5	Entropy as Independence Count	4
6	The Arrow of Time	5
7	Gravity as Dense Shared History	5
8	Expansion as Dilution of Shared History	6
9	Measurement as Local Decoherence	6
10	Black Holes as Decoherence Engines	7
11	The ER = EPR Contact	8
12	The Initial State	8
13	Computational Support: The Gate 1 Ladder	8
13.1	Gate 1A — Total-Spin-Closed Shared Root	8
13.2	Gate 1B1 — Scalar Boundary-Write Tensor	9
13.3	Gate 1B2 — Scalar Orientation Twirl	9
13.4	Gate 1C — Source-Spin Selector	9
13.5	Gate 1 Summary	10
14	Computational Support: The Gate 2 Decoherence Scaffold	10
15	What This Note Establishes	11
16	What This Note Does Not Establish	11
17	Specific Gates for Future Development	12
18	Conclusion	12

1 The Two-Layer Architecture

The V11.2 program established a sharp architectural distinction through computation, not assumption.

Local dent mechanics — the retarded lag residuals that particles move through — produce single-particle quantum phenomena. Interference fringes with the correct de Broglie spacing (V11.2A, H). Which-path decoherence with continuous visibility (V11.2E). Single-slit uncertainty satisfying $\Delta x \cdot \Delta p \geq \hbar/2$ (V11.2F). Trajectory-level guidance (V11.2D). Born equivariance (V11.2G) and a mixing-linked relaxation mechanism (V11.2M3).

But local dent mechanics produce $S_{\text{CHSH}} = 2.000$ exactly — the classical Bell bound [1] (V11.2J). They cannot produce Bell-violating correlations. This is not a failure of the toy implementation. It is a theorem-level result: any model where correlations propagate locally through the lag field at c will obey Bell’s inequality.

The boundary singlet kernel produces $S_{\text{CHSH}} = 2\sqrt{2}$ — the Tsirelson bound (V11.2J). It maintains perfect no-signaling (V11.2L). It is generated by the V11.9 spin-holonomy machinery (V11.2K). And it is invariant under measurement ordering (V11.2L). These correlations have been confirmed experimentally with all major loopholes closed [2].

The architecture is therefore not a choice. It is forced by the computational results:

Phenomenon	Layer	Evidence
Interference	Local dent mechanics	V11.2A
Decoherence	Local dent mechanics	V11.2E
Uncertainty	Local dent mechanics	V11.2F
de Broglie wavelength	Local dent mechanics	V11.2H
Born equivariance	Local dent mechanics	V11.2G
Bell correlations	Boundary grammar	V11.2J
No-signaling	Boundary grammar	V11.2J, L
Measurement-order invariance	Boundary grammar	V11.2L

The question V11.2 left open is: what IS the boundary grammar, physically? Why does entanglement live on the boundary and not in the local geometry?

2 The Proposal: Shared Boundary Roots

The SFT boundary is the accumulated history of all irreversible operations. Every Planck area on it records one completed tick. The boundary does not erase old entries. It only grows. Every entry is permanent.

When a single particle ticks, it writes one entry to the boundary — one Planck area, one history increment, one dN . That entry belongs to that particle. It records what happened. It is that particle’s contribution to the universe’s history.

Now consider pair creation. An electron-positron pair is produced from a photon. At the moment of creation, there is one event, not two. One photon became two particles. One history increment — one boundary entry — records the creation of a correlated pair.

The proposal of this note is that entanglement is the persistence of that shared entry:

Two entangled particles are particles whose boundary histories share a common root entry that has not been resolved into two independent entries.

The particles separate in space. They move to different locations. They interact with different environments. Their worldlines diverge. But their boundary histories still converge backward to the same root — the creation event, recorded as one entry, still intact on the boundary.

The correlations between the particles are not carried through space. They are not encoded in predetermined local properties. They are encoded in the topology of the boundary — the fact that two apparently independent worldlines trace back to one shared entry.

3 Why This Survives Bell’s Theorem

Bell’s theorem [1] rules out local hidden variables — predetermined answers carried independently by each particle through space. The shared boundary root is not a local hidden variable. It is not carried by either particle. It is not located at either particle’s position. It is on the boundary, which is the accumulated history of the universe, not a location in space.

The correlations are not transmitted between the particles at any speed. They are not transmitted at all. They are structural — they exist because the boundary entry has not been factored into independent components. Measuring one particle partially resolves the shared entry. The resolution constrains what the other particle’s measurement can yield. Not through a signal. Through the structure of the entry itself.

No-signaling is maintained because the shared entry constrains the *joint* statistics without constraining the *marginals*. Each particle’s local measurement outcomes, viewed independently, look random. Only the correlations between them reflect the shared root. This is exactly the structure V11.2J–L confirmed computationally: perfect Bell violation with zero marginal deviation.

4 Decoherence as Root Resolution

If entanglement is a shared boundary root, then decoherence is the progressive resolution of that root into independent entries.

Every interaction between an entangled particle and an environmental particle partially resolves the shared entry. The environmental particle gets woven into the boundary structure. The original two-particle root becomes a three-particle root, then a four-particle root, then a root shared with so many environmental particles that no local measurement on the original pair can detect the correlation. The shared structure hasn’t been destroyed. It has been *dispersed* into the environment — diluted across so many boundary entries that it becomes operationally inaccessible.

This is the standard quantum decoherence story, stated in boundary language. But the boundary language adds something the standard story doesn’t: a specific physical substrate for the decoherence process. The shared root is a specific topological feature of the boundary. Its resolution is a specific change in that topology — one entry becoming many. The decoherence rate is the rate at which environmental interactions split shared entries into independent ones.

5 Entropy as Independence Count

In SFT, the total entropy of the boundary is the total number of independent entries:

$$S = \frac{A}{4\ell_P^2}. \tag{1}$$

If decoherence converts shared entries into independent entries, then decoherence *is* entropy growth. Not metaphorically. Operationally. The number of independent boundary entries increases when shared roots are resolved. That increase is measured as entropy.

The second law of thermodynamics — entropy never decreases — becomes the statement that shared roots only resolve, never re-form spontaneously. You cannot un-decohere without doing at least as much work as the decoherence released. Re-entangling two decohered particles would require reassembling the shared boundary root from the dispersed environmental entries — a thermodynamic impossibility for macroscopic systems.

$$\boxed{\text{Decoherence} = \text{resolution of shared entries into independent entries} = \text{entropy growth.}} \quad (2)$$

This identification is not a redefinition. It is a statement that two processes previously treated as separate — quantum decoherence and thermodynamic entropy increase — may be two descriptions of one boundary-topological process.

6 The Arrow of Time

If entropy is the count of independent boundary entries, and entropy always increases, then the arrow of time is the direction of increasing independence.

The early universe was one entry. Maximal sharing. Minimal independence. No classical world. No separate objects. No definite outcomes. Just one unified boundary structure.

As the universe evolved, interactions created new shared roots and resolved old ones. Each resolution increased the number of independent entries. The universe became progressively more classical — more independent, more definite, more separated — as the boundary accumulated independent history.

The arrow of time is not imposed from outside. It is the geometric fact that shared boundary roots can resolve into independent entries but independent entries cannot spontaneously merge back into shared roots. The process is asymmetric because one direction increases the number of entries (resolution) and the other would decrease it (spontaneous re-entanglement), and the boundary only grows.

This connects directly to V11.6's boundary-time duality. Time is the accumulation of irreversible history. The arrow of time is the direction in which that accumulation runs. And the accumulation is, at its deepest level, the progressive resolution of shared boundary structure into independent entries.

7 Gravity as Dense Shared History

A massive object is a concentration of tickers. Many particles, many ticks, many boundary entries, all in a small region. Those particles interact with each other constantly — collisions, electromagnetic interactions, nuclear forces. Every interaction is a shared event. A shared boundary entry. The particles within a massive object are *densely interconnected* through shared recent history.

In the language of this note, a massive object is a region of dense shared boundary structure. The particles inside it have thick, fresh, heavily reinforced shared roots with each other.

The gravitational field of a massive object is, in this picture, the geometric expression of that dense shared structure. Spacetime is more connected near a massive object because the boundary has more shared entries there. A test particle falls toward the mass because it is falling toward

a region of denser connectivity — more shared history, more geometric stitching, more boundary structure per unit volume.

This is a restatement of SFT’s existing gravity result (the lag field gradient from maintenance concentration) in entanglement language. It does not add a new prediction. But it connects gravity to the same mechanism that produces entanglement: shared boundary structure. Gravity is shared roots at macroscopic scale. Entanglement is shared roots at quantum scale. Same mechanism, different regime.

8 Expansion as Dilution of Shared History

Globally, the universe is expanding. In standard SFT, expansion is the boundary growing to accommodate the total ticking demand. In the language of this note, expansion is the *dilution* of shared boundary structure.

As the universe expands, the average distance between particles increases. Their interaction rate decreases. Their shared roots become less dense. New independent history is written continuously, diluting the shared fraction. Distant regions of the universe have less shared boundary structure than nearby regions — not because shared roots were destroyed, but because independent history has been piled on top of them.

Distance, in this picture, is a measure of how much independent history separates two locations. Nearby points share a lot of recent history. Distant points share very little. The expansion of the universe is the process by which independent history accumulates between all locations, progressively diluting the shared fraction.

Phenomenon	Shared boundary interpretation
Gravity	Dense shared history resisting separation
Expansion	Dilution of shared history
Entanglement	Fresh shared root not yet resolved
Decoherence	Progressive resolution of shared into independent
Entropy	Count of independent entries
Arrow of time	Direction of increasing independence
Temperature	Rate of decoherence / root resolution
Measurement	Local decoherence by detector entropy
Classicality	Accumulated decoherence making roots unmeasurable

9 Measurement as Local Decoherence

The measurement problem asks: how does a quantum superposition become a definite outcome? In this framework, the answer is: measurement is local decoherence caused by the detector’s overwhelming independent history.

A detector is a massive object — $\sim 10^{23}$ atoms, each one ticking independently, each one with its own boundary history. When a quantum system interacts with the detector, the system’s remaining shared roots are instantly overwhelmed by the detector’s independent structure. The system’s correlations with anything else get drowned in the detector’s entropy.

An order-of-magnitude estimate makes the asymmetry vivid. Each detector atom at room temperature has a thermal energy of roughly $k_B T \approx 4 \times 10^{-21}$ J. Its Compton ticking rate is $mc^2/h \approx 10^{23}$ Hz for a typical atomic mass. In one interaction timescale, a single detector atom contributes roughly $k_B T/(\hbar\omega_C) \sim 10^{12}$ independent boundary entries to the interaction record. Multiplied by $\sim 10^{23}$ atoms in a macroscopic detector, the detector contributes approximately 10^{35} independent entries per measurement interaction. The quantum system’s shared root is *one* entry. The ratio of independent to shared is $\sim 10^{35}$. The shared root is outnumbered by thirty-five orders of magnitude.

The outcome is definite because the detector’s boundary history is so massive that the system’s shared roots are completely resolved by the interaction. The “collapse” is not a mysterious nonlocal event. It is the local boundary-topological consequence of a small shared structure interacting with an enormous independent structure.

This also explains why macroscopic objects appear classical. A baseball has $\sim 10^{26}$ atoms whose internal interactions have been decohering their external correlations since the atoms were forged in a supernova. All external shared roots have been buried under an astronomical amount of independent internal history. The baseball looks classical because it has been maximally decohered by its own internal dynamics.

Conversely, quantum effects become visible at low temperatures — superfluidity, superconductivity, Bose-Einstein condensation — because the decoherence rate drops below the rate at which shared roots maintain themselves. The particles can sustain coherent shared structure because the environmental noise is too weak to resolve it.

10 Black Holes as Decoherence Engines

A black hole has entropy proportional to its horizon area: $S_{\text{BH}} = A/(4\ell_P^2)$. This is an enormous number — $\sim 10^{77}$ for a solar-mass black hole, more than all the stars in the observable universe combined.

In the shared-boundary picture, a black hole is a decoherence engine. Every particle that falls behind the horizon has its external shared roots *severed* by the horizon. The horizon is a decoherence wall — it cuts the boundary connections between the infalling matter and the outside universe in one event.

The BH entropy counts the severed roots — the total number of external correlations that have been resolved by the horizon’s formation and growth. Each Planck area on the BH horizon is one severed shared entry, now independent, now contributing to the BH’s private boundary rather than the cosmological boundary.

And Hawking radiation is the slow reverse process. The BH boundary leaks its entries back to the cosmological boundary, one bit at a time, at a rate set by the BH temperature. The information is not destroyed. It migrates from one accounting surface to another.

This connects directly to the V11.10 late-time result. The late-time crossing at $a \approx 0.47$ – 0.50 is carried by mass plus black-hole area-growth response, with the BH channel providing a delayed boundary-area accumulation that shapes the dominance-gradient handoff. In the language of this note, the BH channel in V11.10 is the cosmological signature of the universe’s most efficient decoherence engines processing shared structure into concentrated independent boundary area. The late-time acceleration is not driven by a cosmological constant. It is driven by the accumulated decoherence footprint of the universe’s black hole population.

11 The ER = EPR Contact

In 2013, Maldacena and Susskind conjectured that every entangled pair is connected by a wormhole (an Einstein-Rosen bridge), and every wormhole connects an entangled pair [3]. This conjecture, called ER = EPR, proposes that the connectivity of spacetime is built from entanglement.

The shared-boundary-root picture provides a natural SFT reading of ER = EPR. The shared root IS the connection between the two particles. It is not a tunnel through space. It is a single boundary entry that serves two spatial locations. The “bridge” is the topological fact that two worldlines converge to one entry when traced backward through the boundary’s history.

Van Raamsdonk’s 2010 result strengthens this connection [4]. He showed within the AdS/CFT framework that reducing entanglement between two boundary regions causes the corresponding bulk spacetime to develop a tear. Remove all entanglement and spacetime disconnects entirely. His conclusion: spacetime is stitched together by entanglement.

In SFT language: spacetime is stitched together by shared boundary structure. Remove the shared entries and the geometric connectivity dissolves. The fabric of spacetime is the coarse-grained expression of the density of shared boundary roots.

12 The Initial State

Every particle’s worldline traces backward through time to a creation event. Follow the creation events backward far enough and they all converge — to the initial state, the one unified process before the shatter.

At the beginning, there was one entry. One boundary. One shared root. Not many entangled particles. One thing. The concept of “entanglement” requires two systems. Before the shatter, there was one system. The word “entangled” does not apply. The word “one” applies.

The shatter was the first resolution event. One entry began splitting into many. Shared structure began resolving into independent structure. The universe began its long progression from maximum sharing to increasing independence. Every subsequent tick, every interaction, every decoherence event, every entropy increase has been a continuation of that original resolution.

The classical world is the accumulated result of $\sim 10^{122}$ ticks of progressive disentanglement from the original unity. The quantum world — entanglement, coherence, interference — is the surviving remnant of shared structure that hasn’t been fully resolved yet.

13 Computational Support: The Gate 1 Ladder

The conceptual proposal above was subjected to a four-gate computational ladder testing whether the shared-root picture can produce the singlet record rather than merely import it. The results are summarized here; full numerical outputs are in the companion data files.

13.1 Gate 1A — Total-Spin-Closed Shared Root

Gate 1A tested the minimal closure condition for a shared two-spinor root:

$$J_{\text{total}}^i |\text{root}\rangle = 0, \quad i = x, y, z. \quad (3)$$

The nullspace solution was unique (nullity = 1). The derived root was:

$$|\text{root}\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \quad (4)$$

Projection onto arbitrary measurement axes recovered $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$ with RMS error 2.1×10^{-16} , $S_{\text{CHSH}} = 2\sqrt{2}$ to 4.4×10^{-16} , marginal deviation 2.2×10^{-16} , and measurement-order delta exactly zero.

Controls: product states ($S = 1.41$), classical mixtures ($S = 1.41$), the triplet $m = 0$ state ($S = 0$), and local hidden-variable models ($S = 2.00$) all failed to reproduce the singlet kernel.

Claim: A single total-spin-closed shared root uniquely yields the singlet record.

13.2 Gate 1B1 — Scalar Boundary-Write Tensor

Gate 1B1 moved one step deeper by treating pair creation as a scalar boundary write with two outgoing spinor legs. The $\text{SU}(2)$ -invariance condition on a two-spinor write tensor A_{ab} :

$$G_i A + A G_i^T = 0, \quad G_i = \sigma_i/2, \quad (5)$$

has a one-dimensional nullspace. The unique invariant tensor is proportional to the antisymmetric epsilon tensor:

$$A_{ab} \propto \epsilon_{ab} \quad \longleftrightarrow \quad \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \quad (6)$$

A scalar two-spinor boundary write does not merely allow the singlet. It selects the singlet as the unique $\text{SU}(2)$ -invariant two-spinor tensor.

Claim: Scalar symmetry of the boundary write uniquely selects the epsilon tensor, which is the singlet.

13.3 Gate 1B2 — Scalar Orientation Twirl

Gate 1B2 tested the isotropy of the scalar write. If the boundary write carries no preferred spin axis, then a generic two-leg write should be averaged over $\text{SU}(2)$ orientations:

$$T[A] = \int dU U A U^T. \quad (7)$$

The result is an orientation filter. Non-invariant components are suppressed. The only surviving two-spinor component is the antisymmetric epsilon component. An oriented product seed, after $\text{SU}(2)$ twirling and normalization, reproduces the singlet diagnostics: $E(\mathbf{a}, \mathbf{b}) \approx -\mathbf{a} \cdot \mathbf{b}$ and $S_{\text{CHSH}} \approx 2\sqrt{2}$. Symmetric triplet and delta-tensor seeds do not survive as scalar invariant components.

Claim: A scalar/isotropic boundary write acts as an $\text{SU}(2)$ -invariant projector. The singlet is the component that survives loss of preferred orientation.

13.4 Gate 1C — Source-Spin Selector

Gate 1C was a claim-discipline audit. The shared-root picture must not imply that every pair-creation process produces a singlet. The source representation matters.

The decomposition $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ separates the scalar singlet sector from the vector triplet sector. The scalar projector selects the singlet with $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$ and $S_{\text{CHSH}} = 2\sqrt{2}$. The isotropic vector triplet mixture gives $E_{\text{triplet}}(\mathbf{a}, \mathbf{b}) = +\frac{1}{3} \mathbf{a} \cdot \mathbf{b}$, with no-signaling marginals but without the singlet Bell kernel.

Claim: Source quantum numbers determine which boundary record is written. Scalar shared roots select the singlet. Vector shared roots select triplet structure. The framework is specific, not universal.

13.5 Gate 1 Summary

Gate	Question	Result
1A	Does spin closure of a shared root yield the singlet?	Pass
1B1	Does a scalar two-spinor write select ϵ_{ab} ?	Pass
1B2	Does isotropic SU(2) twirling project onto ϵ_{ab} ?	Pass
1C	Do source quantum numbers select singlet vs triplet?	Pass

The derived chain, with no imported quantum mechanics:

$$\boxed{\text{scalar boundary write} \Rightarrow \epsilon_{ab} \Rightarrow |\text{singlet}\rangle \Rightarrow E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b} \Rightarrow S_{\text{CHSH}} = 2\sqrt{2}} \quad (8)$$

14 Computational Support: The Gate 2 Decoherence Scaffold

Gate 1 showed that the shared-root picture can *create* the correct entanglement record. Gate 2 tested whether it can correctly *destroy* it through environmental interaction — whether the standard decoherence-to-classicality chain maps onto boundary-root resolution.

Twenty-four sub-gates (2A through 2X) were tested, covering per-interaction root resolution, time-domain rate laws, multi-channel kinetics, Gaussian marker overlap, environmental benchmarking, Lindblad/Kraus channel equivalence, pointer-basis stability, quantum Darwinism (redundant boundary records), Born-weight preservation, recoherence/eraser boundaries, classical-limit scaling, thermodynamic Landauer bookkeeping, visibility-distinguishability complementarity ($D^2 + V^2 = 1$), no-signaling and measurement-order invariance, a Born-weight symmetry scaffold (equal-weight symmetry, rational fine-graining, continuity extension, coarse-graining additivity, POVM/Naimark consistency), Born-exponent uniqueness ($q = 2$ selected by fine-graining), adversarial negative controls, and a synthesis manifest. All twenty-four sub-gates passed.

The combined root-resolution law is:

$$\boxed{\kappa(t) = \exp \left[-t \sum_i n_i \sigma_i v_i \left(1 - \exp \left[-\frac{\Delta x^2}{8\sigma_{q,i}^2} \right] \right) \right]}, \quad (9)$$

where κ is the accessible shared-root coherence, n_i , σ_i , v_i are the density, cross-section, and velocity of environmental marker species i , Δx is the spatial separation of the superposition branches, and $\sigma_{q,i}$ is the Gaussian position spread of marker species i . For the two-branch singlet seed, $C(t) = \kappa(t)$, $S_{\text{root}}(t) = H_2\left(\frac{1+\kappa(t)}{2}\right)$, and $S_{\text{CHSH}}^{\text{max}}(t) = 2\sqrt{1 + \kappa(t)^2}$.

Gate 2 establishes that, once a shared boundary root exists, the standard decoherence-to-classicality machinery is recovered in boundary-ledger language: environment marks reduce accessible coherence, generate classical records, preserve causality, respect thermodynamic bookkeeping, and support the standard Born-weight scaffold with $q = 2$ uniquely selected by fine-graining consistency.

Gate 2 does *not* derive the microscopic root-resolution interaction from the SFT action. It does not derive the environmental channel parameters from first principles. It does not claim a named calibrated experimental fit. It does not derive the Born rule from SFT dynamics — the scaffold imports symmetry, additivity, and continuity as explicit assumptions. These open items are documented in the companion addendum.

The full Gate 2 scaffold, with all twenty-four sub-gate reports, numerical outputs, and detailed derivations, is documented in the companion document: *SFT V11.12 Gate 2 Addendum: Decoherence as Boundary-Root Resolution*.

15 What This Note Establishes

This note establishes a conceptual framework with computational support across two gate ladders. Specifically:

1. Entanglement can be interpreted as shared boundary structure — a single history entry serving two spatial locations.
2. This interpretation is consistent with V11.2’s two-layer architecture: local dent mechanics cannot produce Bell violation ($S = 2$), while boundary-level shared entries can ($S = 2\sqrt{2}$).
3. A scalar shared boundary write uniquely produces the singlet record through $SU(2)$ invariance. The Bell kernel $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$ emerges from projection of the derived root, not from an imported probability rule (Gates 1A–1B2).
4. Source quantum numbers determine which record is written. Scalar sources produce singlets; vector sources produce triplets (Gate 1C).
5. Decoherence, entropy growth, and the arrow of time can be unified as the progressive resolution of shared boundary entries into independent entries.
6. Gravity and expansion can be restated as dense and dilute shared boundary structure, respectively, connecting to existing SFT results in new language.
7. Measurement can be interpreted as local decoherence from detector entropy, with an estimated $\sim 10^{35}$ -fold asymmetry between detector independent entries and the system’s shared root.
8. Black holes can be interpreted as concentrated decoherence engines whose entropy counts severed shared roots, connecting to V11.10’s BH area-growth channel in the late-time crossing.
9. The framework makes contact with the $ER = EPR$ conjecture [3] and Van Raamsdonk’s entanglement/spacetime result [4].
10. The standard decoherence-to-classicality chain maps onto boundary-root resolution, with twenty-four sub-gates passing across rate laws, channel equivalence, pointer-basis stability, quantum Darwinism, Born-weight preservation, thermodynamic bookkeeping, complementarity, no-signaling, and Born-exponent uniqueness (Gate 2A–2X). The assumptions underlying the Born-weight scaffold — symmetry, additivity, continuity — are explicit.

16 What This Note Does Not Establish

The following are explicitly not claimed:

1. A derivation of the Born rule from SFT dynamics. Gate 2 provides a Born-weight scaffold where $q = 2$ is uniquely selected by fine-graining consistency, but the scaffold imports symmetry, additivity, and continuity as explicit assumptions rather than deriving them from the SFT action.
2. A derivation of the scalar boundary-write condition from the microscopic SFT action. Gates 1A–1C show that IF the write is scalar, THEN the singlet follows uniquely. The “if” is not yet connected to the Lagrangian.
3. A proof of $ER = EPR$. The shared-boundary picture is consistent with $ER = EPR$ but does not constitute a proof.
4. A derivation of decoherence rates from the microscopic SFT action. Gate 2 provides a quantitative root-resolution rate law that matches the standard collisional-decoherence benchmark,

but the environmental channel parameters (n, σ, v, σ_q) are imported from kinetic theory, not derived from SFT dynamics.

5. A derivation of the Bekenstein-Hawking entropy from shared-root severing. The interpretation is proposed, not derived.
6. A proof that spacetime connectivity equals boundary sharing density. The connection to Van Raamsdonk's result is noted, not demonstrated within SFT.
7. A replacement for standard quantum decoherence theory. The standard theory works. This note proposes a physical substrate for the mathematical process it describes.
8. A mechanism for producing the initial shared state. The "one entry at the beginning" is a boundary condition, not a dynamical result.

17 Specific Gates for Future Development

Gate 1B_{action} (open): Derive the scalar boundary-write condition from the SFT Lagrangian. Show that the repaired potential $V_0(e^{-\psi} - 1)^2$, being a scalar, produces scalar boundary writes for pair-creation processes.

Gate 2 (scaffold closed, action derivation open): The decoherence-to-classicality scaffold passes with twenty-four sub-gates (2A–2X). The root-resolution rate law matches the collisional-decoherence benchmark. Born-weight uniqueness ($q = 2$) is selected by fine-graining. Remaining: derive the microscopic root-resolution interaction from the SFT action, derive the environmental channel parameters from first principles, and compare against a named calibrated decoherence experiment. Full scaffold documented in the Gate 2 companion addendum.

Gate 3 (open): BH entropy from shared-root severing. Derive $S_{\text{BH}} = A/(4\ell_P^2)$ by counting the external shared roots severed when matter falls behind a horizon. If the counting doesn't produce the Bekenstein-Hawking formula, the decoherence-engine interpretation fails.

Gate 4 (open): Spacetime connectivity from boundary sharing. Show that the coarse-grained density of shared boundary entries produces a metric-like structure whose geodesics match the SFT lag-field geometry. If the shared structure doesn't reproduce the correct spacetime geometry, the entanglement-gravity connection fails.

Gate 5 (open): Born rule from SFT dynamics. The Gate 2 scaffold selects $q = 2$ under explicit assumptions of symmetry, additivity, and continuity. This gate requires deriving those assumptions from the SFT action or from boundary-root dynamics directly. V11.2M3's mixing-linked relaxation result ($H_f/H_0 = 0.39$, perfect mixing-relaxation correlation) provides a numerical seed.

18 Conclusion

The universe began as one shared entry. Every tick since then has been one step in the resolution of that shared structure into independent components. Entanglement is the surviving shared structure. Decoherence is the ongoing resolution. Entropy counts the resolved entries. Time is the direction of resolution. Gravity is where shared structure is dense. Expansion is where it is dilute. Measurement is where a detector's enormous independent history overwhelms a system's remaining shared roots.

These are not seven separate ideas bolted together. They are seven projections of one process: the progressive independence of the universe’s boundary structure, driven by irreversible ticking at the rate of one operation per operation.

A four-gate algebraic ladder shows that the singlet spine holds: a scalar boundary write, through $SU(2)$ invariance alone, uniquely produces the singlet record that generates the Bell correlations. A twenty-four-gate decoherence scaffold shows that the resolution spine holds: environment marks reduce shared-root coherence at the standard collisional rate, generate classical records, preserve causality, and select $q = 2$ as the unique Born exponent. The singlet is not imported. The decoherence law is not imported. Both emerge from the shared-root picture.

The strongest statement this note makes is also the simplest:

Entanglement is not a strange quantum phenomenon that defies classical intuition. Entanglement is the original state. Classical independence is the anomaly — the hard-won result of $\sim 10^{122}$ ticks of progressive boundary resolution. The question is not “why are things entangled?” The question is “why did things become separate?” And the answer is: because existence has a cost, the cost is paid in boundary entries, and every entry resolves one more piece of the original unity into independence.

References

- [1] J. S. Bell, “On the Einstein Podolsky Rosen Paradox,” *Physics* **1**, 195–200 (1964).
- [2] A. Aspect, J. Dalibard, G. Roger, “Experimental Realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment: A New Violation of Bell’s Inequalities,” *Physical Review Letters* **49**, 1804–1807 (1982).
- [3] J. Maldacena, L. Susskind, “Cool horizons for entangled black holes,” *Fortschritte der Physik* **61**, 781–811 (2013). arXiv:1306.0533.
- [4] M. Van Raamsdonk, “Building up spacetime with quantum entanglement,” *General Relativity and Gravitation* **42**, 2323–2329 (2010). arXiv:1005.3035.

This note originated in a speculative conversation between the author and Claude/Opus (Anthropic) and was developed with computational support from ChatGPT/5.5 (OpenAI). The Gate 1 ladder was implemented and verified by 5.5. Additional input from ChatGPT/5.4 (OpenAI), Gemini/Ender (Google), and Grok (xAI) contributed to earlier stages of the V11 quantum foundations program. The conceptual framework is exploratory. The explicit gates are the theory’s contract with itself.